1. Genome assembly refers to 1 point

- ( ) A computational method to identify the genes being expressed in a cell or tissue
- ( ) The process whereby a cell copies its DNA
- (•) A computational method for reconstructing chromosomes from short reads
- ( ) A computational process to translate highlevel code into machinelevel instructions

2. Which of the following is not true about DNA? 1 point

- ( ) Each strand has a direction
- ( ) It is a doublestranded molecule
- ( ) One strand is complementary to the other
- (•) It doesn't matter which direction you write the sequence in

3. RNA molecules are translated into 1 point

- ( ) DNA molecules
- ( ) Modified RNA molecules
- (•) Proteins
- ( ) Introns

4. Messenger RNA is 1 point

- (•) A template from which proteins are constructed by ribosomes
- ( ) A special signal that helps a cell communicate with other cells
- ( ) A reverse copy of DNA
- ( ) The genetic material inherited by offspring

5. DNA is copied into DNA in order to 1 point

- (•) Replicate a cell
- ( ) Encourage evolutionary changes
- ( ) Respond to an infection
- ( ) Create species diversity

6. Evolutionary biology involves the study of 1 point

- (•) The process of natural selection that allows some DNA mutations to survive and cause others to die out
- ( ) The process through which RNA is exported from the nucleus
- ( ) The origin of the very first living organisms
- ( ) The purpose of life on earth

7. Which of the following can we measure with next generation sequencing? **1 point**

- ● DNA-protein binding
- ○ Cell structure
- ○ Protein levels
- ○ RNA secondary structure

8. What is the first step in ChIP-sequencing to measure protein-DNA binding? **1 point**

- ○ Antibody pulldown of the linked proteinDNA fragments
- ○ Fragmenting the DNA
- ○ Sequencing the bound DNA fragments
- ● Cross-linking proteins to the DNA

9. Which of the following can be measured using bisulfite conversion and then sequencing? **1 point**

- ○ Transcription-factor binding
- ● DNA methylation
- ○ DNA variants
- ○ RNA abundances

10. What is the primary measurement technology used in most modern genomics experiments? **1 point**

- ○ Sanger sequencing
- ○ Nanopore sequencing
- ○ Polymerase chain reaction
- ● Next generation sequencing
- ○ Oligonucleotide arrays
- ○ Western blotting